

Computer Graphics Basics - 2

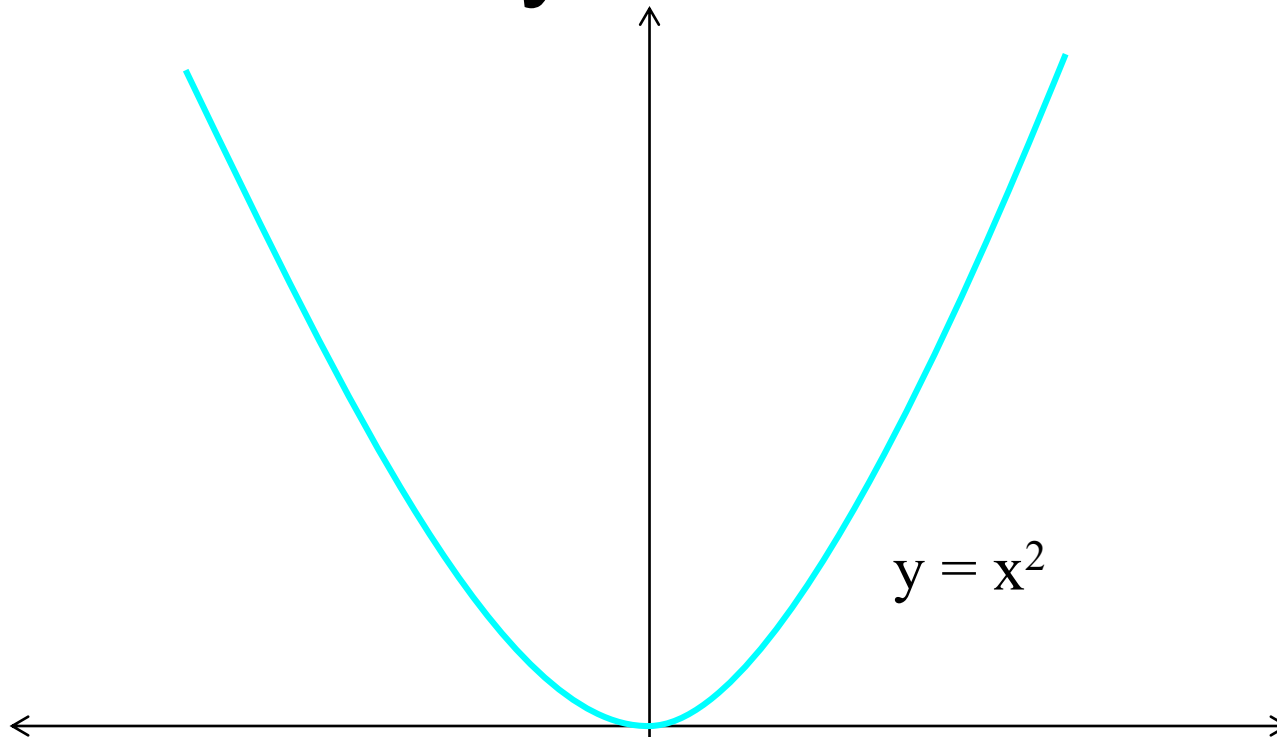
Contents

- Curve and Surface Representation
 - Explicit form
 - Implicit form
 - Parametric form
 - Piecewise form

Curve and Surface Forms

- Explicit
- Implicit
- Parametric
- Piecewise

How would you describe this?



$$y = x^2$$

Explicit representation –
represents a curve with one
variable as a function of one or
more other variables. $y=f(x)$

Explicit Representation: Some Examples

- 2D

$$y = f(x), \text{ eg. } y = mx + c$$

- 3D

$$z = f(x, y), \text{ eg. } z = ax + by + c$$

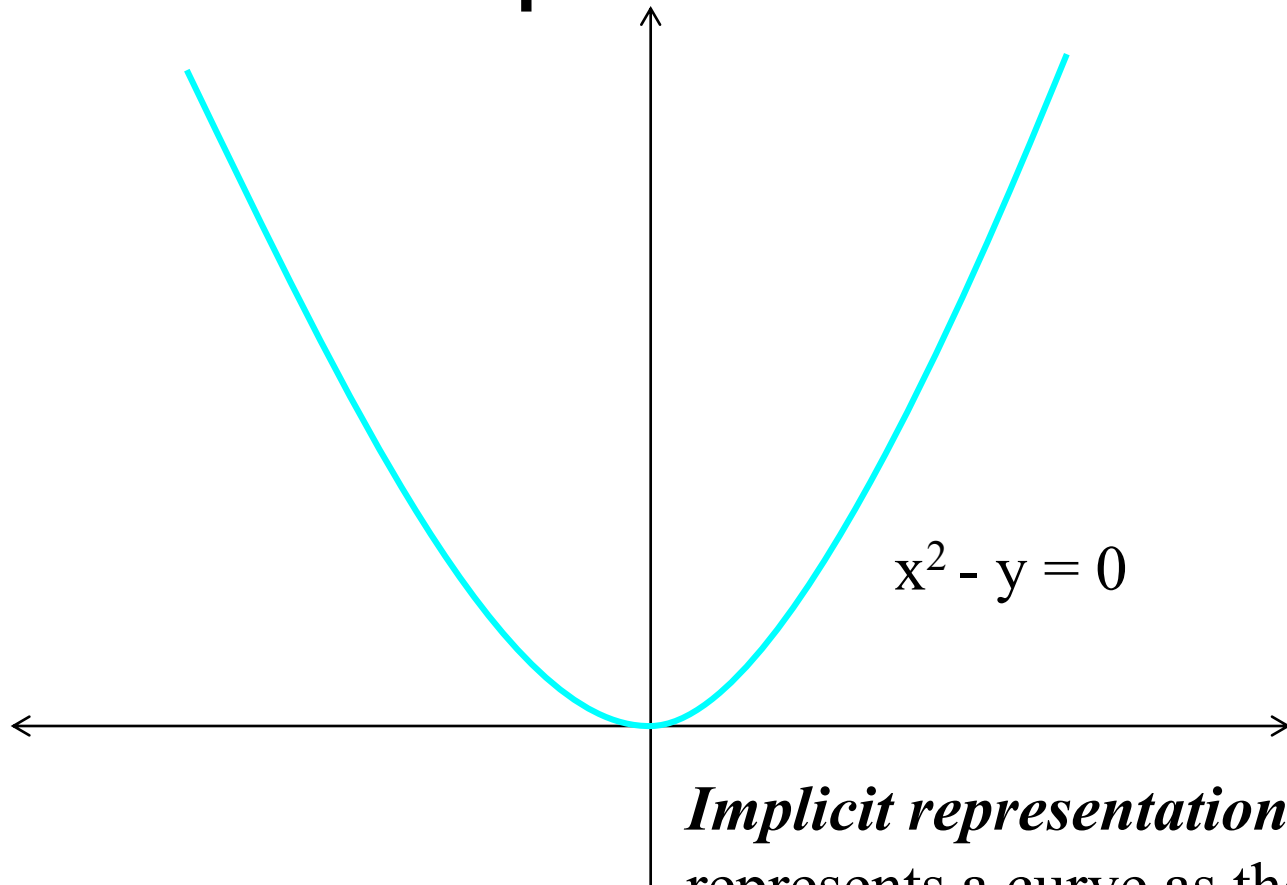
- What's the representation for a 3D line?

3D Line requires two equations: the two planes which intersect in that line.

What is the “ $y = mx + c$ ” form for a vertical line?

The Explicit form falls apart here.

Implicit Form



Implicit representation –
represents a curve as the zero
of a function of all coordinate
variables: $f(x, y) = 0$

Implicit Representation: Some Examples

- 2D

$$f(x, y) = 0$$

$$ax + by + c = 0$$

Line

$$x^2 + y^2 - r^2 = 0$$

Circle

- 3D

$$f(x, y, z) = 0$$

$$ax + by + cz + d = 0$$

Plane

$$x^2/a^2 + y^2/b^2 + z^2/c^2 - r^2 = 0$$

Ellipsoid

Point Classification Using Implicit Form

- The function is really a membership test

$f(x, y, z) = 0 \rightarrow$ Is $f(x, y, z) = 0$ for a point?

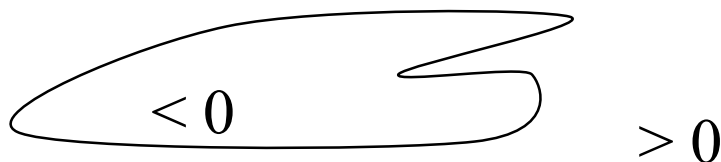
Point classification is easy.

Given any point (x, y, z) ,

If $f(x, y, z) = 0$, then on the surface;

> 0 , then outside/one side of the surface;

< 0 , then inside/other side of the surface;



Problems in Using Implicit Form

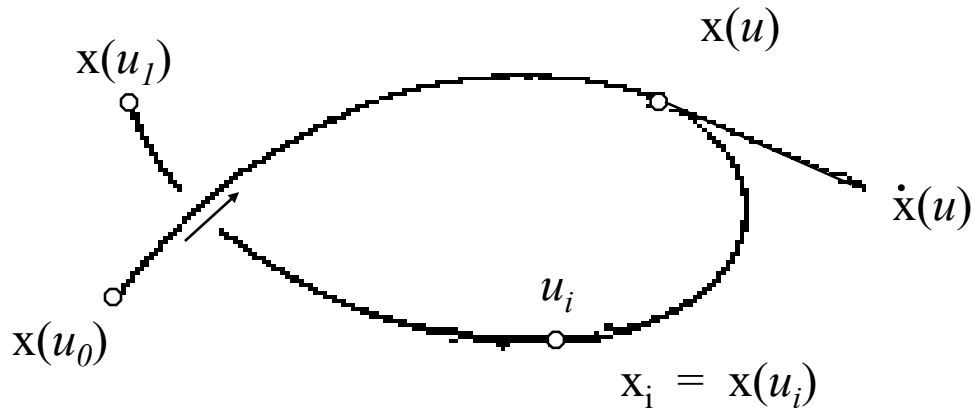
- Point classification is easy.
- But drawing is not.
- Impractical to test every pixel for classification.
- Need a generative form, one that allows us to compute the points on the curve/surface by evaluation of some simple function.

Parametric Representation of Curves

Make x , y , z as separate functions of a parameter.

A planar curve:

$$\begin{aligned}x &= x(u) \\ y &= y(u)\end{aligned} \quad u \in (u_0, u_1)$$



A space curve: $(x(u), y(u), z(u))$

Parametric Form: Some Examples

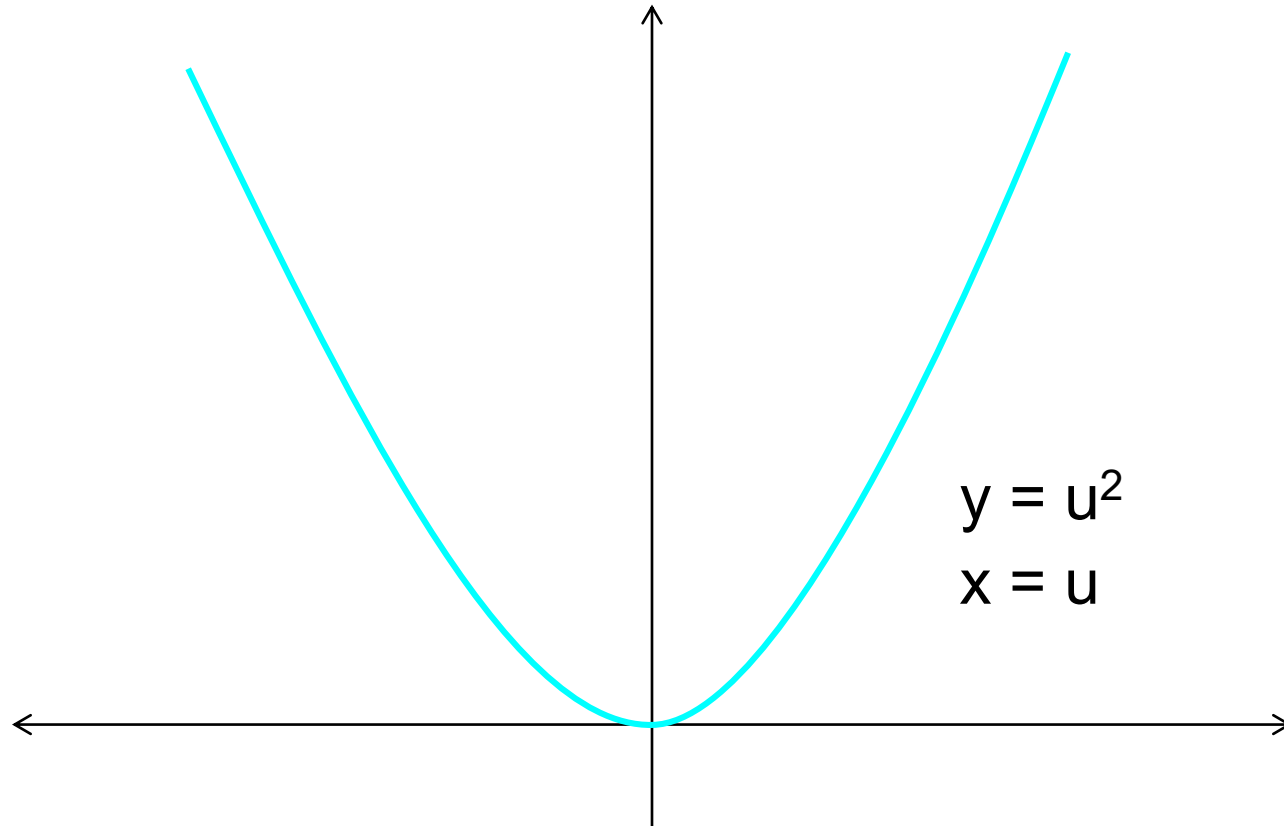
$$\Rightarrow x = x_0 + (x_1 - x_0) * u$$

$$\Rightarrow x = r * \cos(u)$$

$$y = y_0 + (y_1 - y_0) * u$$

$$y = r * \sin(u)$$

Parametric Representation



Parametric Form: More Examples

$$\Rightarrow x = x_u + (x_1 - x_0) * u \quad y = y_0 + (y_1 - y_0) * u$$

$$\Rightarrow x = r * \cos(u) \quad y = r * \sin(u)$$

Functions with two arguments are used to obtain surface definitions.

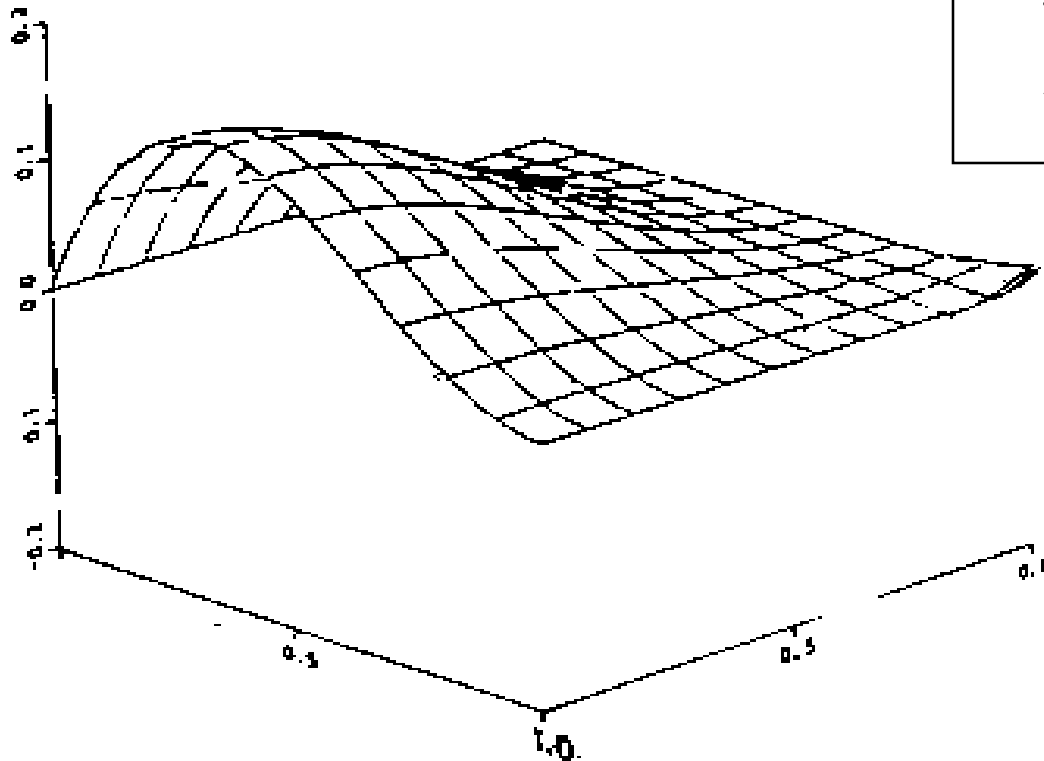
Say, $f(u,v)$ is the function.

Ellipsoid

$$x = a \cos(u) \sin(v)$$

$$y = b \sin(u) \sin(v)$$

$$z = c \cos(v)$$



Curve Tangents

- Obtained by taking first derivative.
- For explicit form : $\frac{dy}{dx}$
- For implicit form take partial derivatives

$$\left(\frac{\delta f}{\delta x}, \frac{\delta f}{\delta y} \right)$$

- For parametric form, take the first derivative of each coordinate function

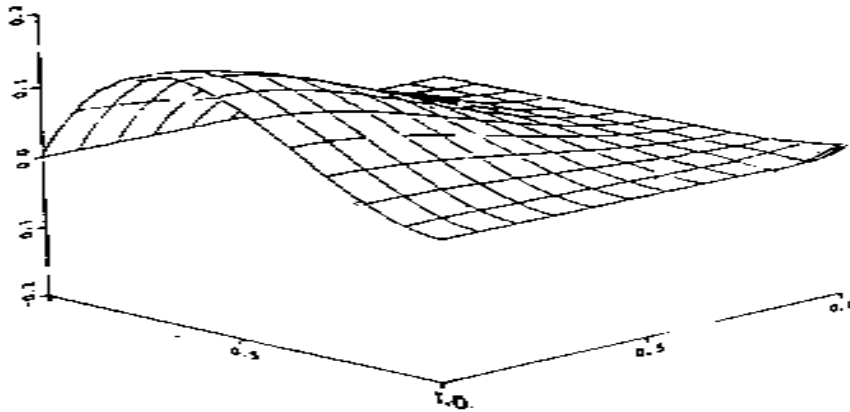
$$\left(\frac{dx}{du}, \frac{dy}{du} \right)$$

Surface Normals

Taking the derivatives of a function with respect to one of the arguments gives us “partial derivatives”. If $f(u,v)$ is the function then

$$\frac{\delta f}{\delta u}, \frac{\delta f}{\delta v}, \frac{\delta^2 f}{\delta u^2}, \frac{\delta^2 f}{\delta v^2}, \frac{\delta^2 f}{\delta u \delta v}$$

etc. are partial derivatives

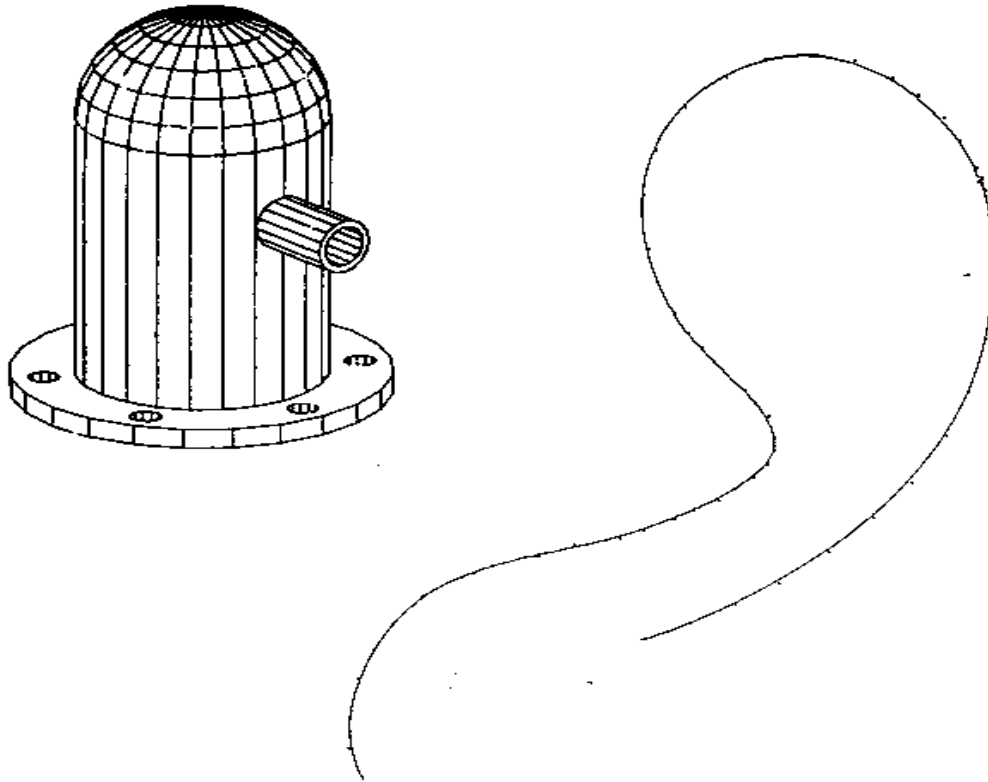


Normal is given by:

$$\left(\frac{\delta f}{\delta u} \times \frac{\delta f}{\delta v} \right)$$

Piecewise Linear Representations

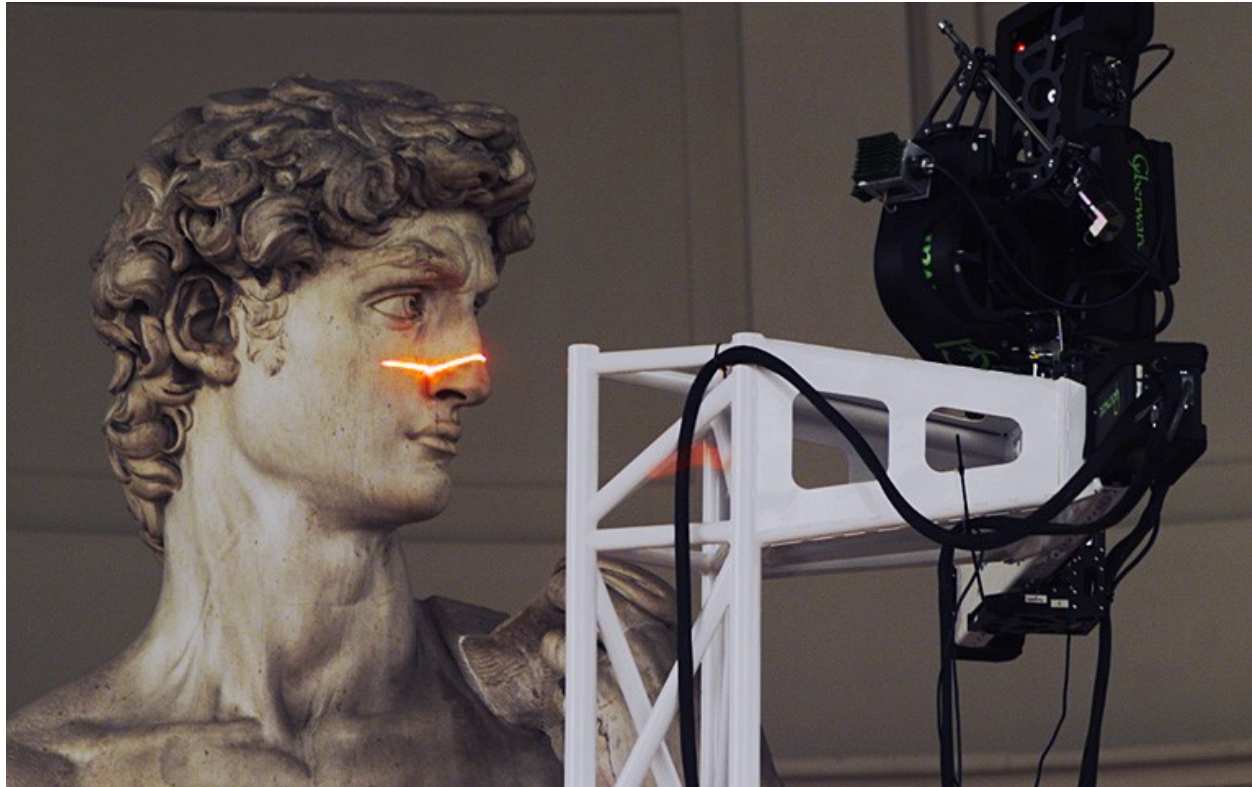
Any curve can be approximated by a connected sequence of line segments. Similarly any surface can be approximated by a connected set of planar polygonal faces.



Meshes from Scanning

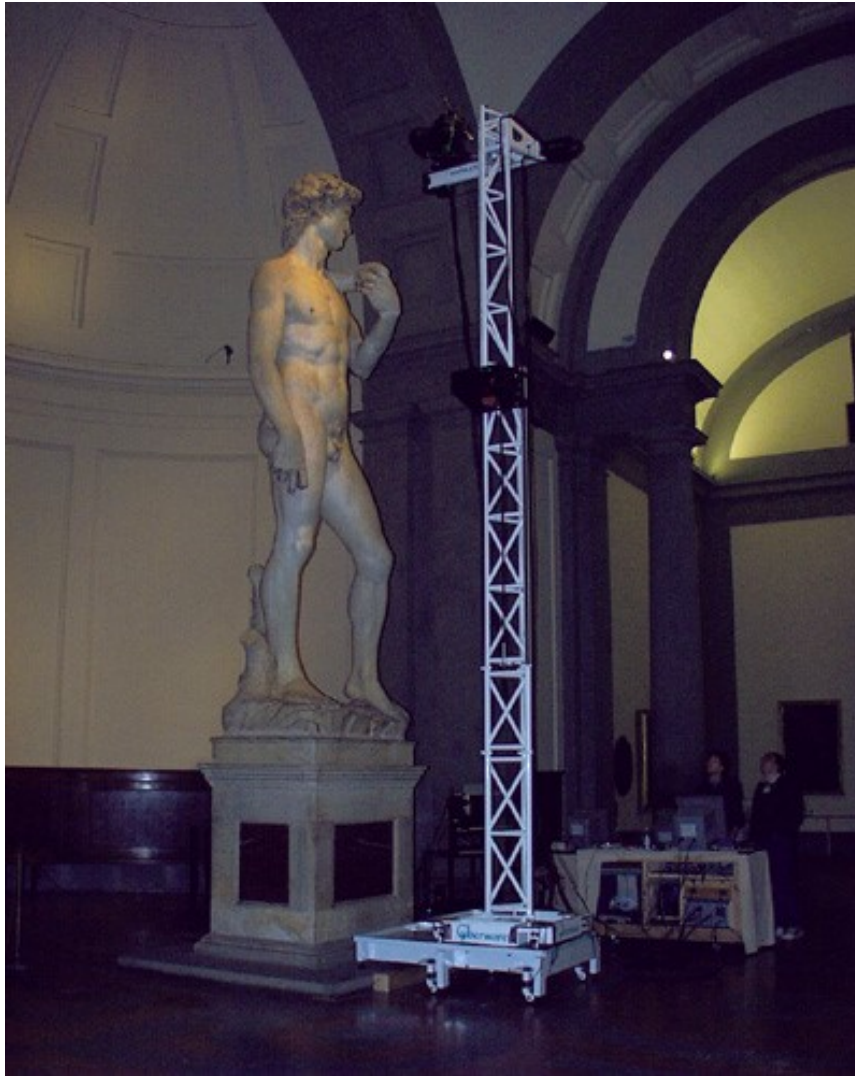
- Laser scanners sample 3D positions
 - One method uses time of flight
 - Some take images also for use as textures
 - Famous example: Scanning the David
- Software then takes thousands of points and builds a polygon mesh out of them
- Research topics:
 - Reduce the number of points in the mesh
 - Reconstruction and re-sampling!

Picture



<http://www-graphics.stanford.edu/projects/mich>

Picture



<http://www-graphics.stanford.edu/projects/mich>

The 3D David

Normal Vectors

- Normal vectors give information about the true surface shape
- Per-Face normals:
 - One normal vector for each face, stored as part of face
 - Flat shading
- Per-Vertex normals:
 - A normal specified for every vertex (smooth shading)

Computing Normal Vectors

- Many models do not come with normals
Models from laser scans are one example
- Per-Face normals are easy to compute:
Normalize cross product of two edges
- For per-vertex normals:
Compute per-face normals
Average normals of faces surrounding vertex
 - A case where neighbor information is useful

End.